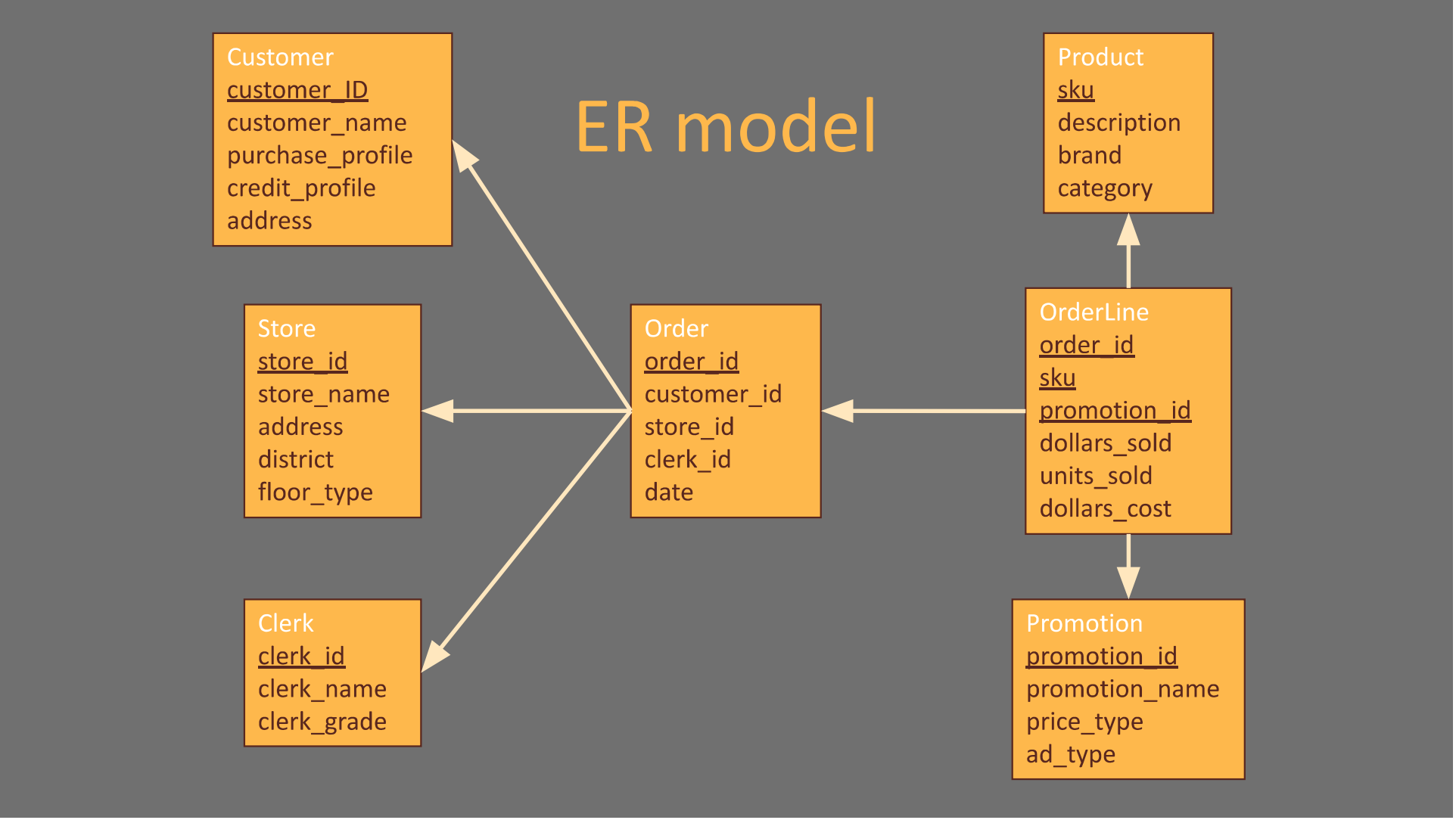
Dimensional Modeling

An alternative to ER modeling

Only an overview.
So, we'll view it through the lens of ER modeling + denormalization.

Emphasizes decision making & use of historical data

- Fast retrieval & aggregation of data – optimized via denormalization
- Less concern with updating data & maintaining consistency while updating


ER model
Customer
customer_ID
customer_name
purchase_profile
credit_profile
address
Product
sku
description
brand
category
Store
store_id
store_name
address
district
floor_type
Order
order_id
customer_id
store_id
clerk_id
date
OrderLine
order_id
sku
promotion_id
dollars_sold
units_sold
dollars_cost
Clerk
clerk_id
clerk_name
clerk_grade
Promotion
promotion_id
promotion_name
price_type
ad_type

Facts & dimensions

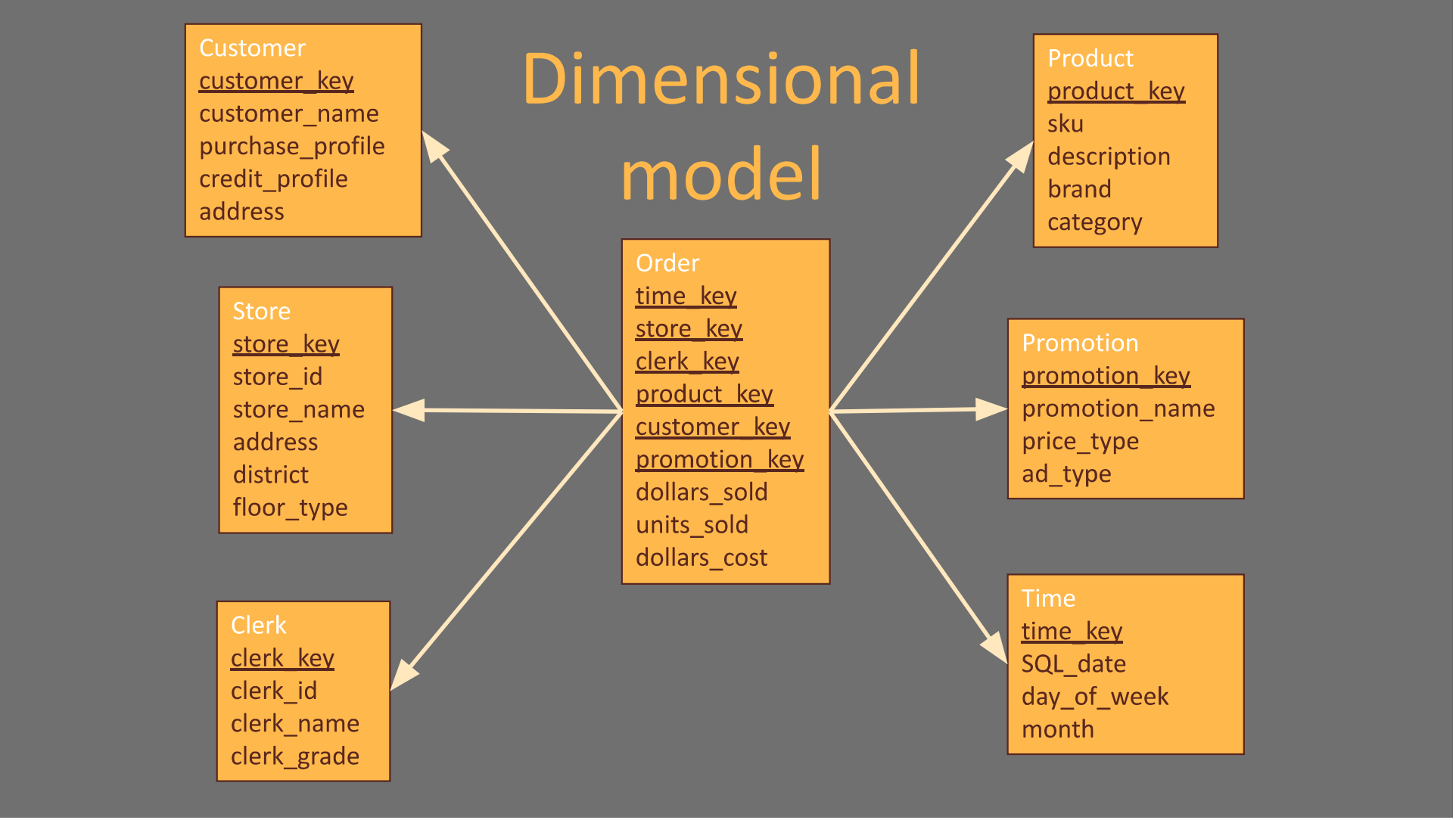
Facts: The data you want to summarize & aggregate

- E.g.: items sold, item sales price
- Data often numeric & additive
  - Additive: units sold, total revenue, number of enrolled students
  - Non-additive: price per unit

Dimensions: The data you want to aggregate over

- E.g., stores, customers, sales people, sales period

Design depends on desired queries.

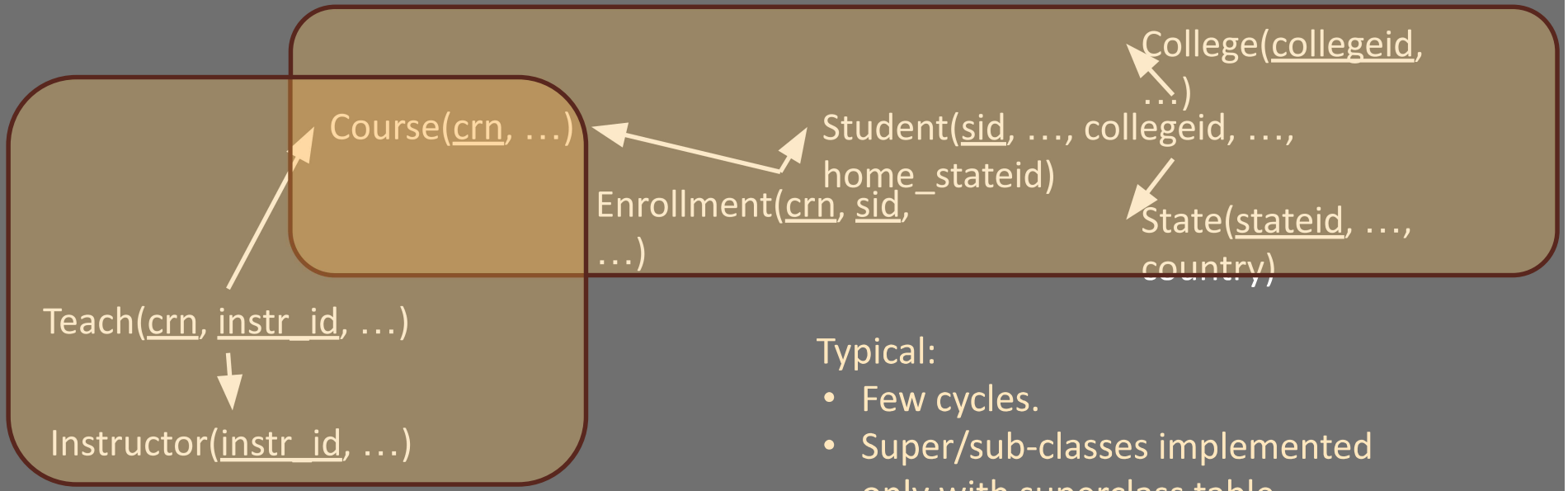

Dimensional model
Customer
customer_key
customer_name
purchase_profile
credit_profile
address
Store
store_key
store_id
store_name
address
district
floor_type
Clerk
clerk_key
clerk_id
clerk_name
clerk_grade
Order
time_key
store_key
clerk_key
product_key
customer_key
promotion_key
dollars_sold
units_sold
dollars_cost
Product
product_key
sku
description
brand
category
Promotion
promotion_key
promotion_name
price_type
ad_type
Time
time_key
SQL_date
day_of_week
month

Standard dimensional modeling

- Identify the potential queries.
- Identify the facts & dimensions used in such queries.
- Group facts & dimensions into tables.
  - Each query should use only one fact table & its dimension tables.

Design often resembles starflakes

Starflake = tree of junction table & child tables in 1-to-many relationships

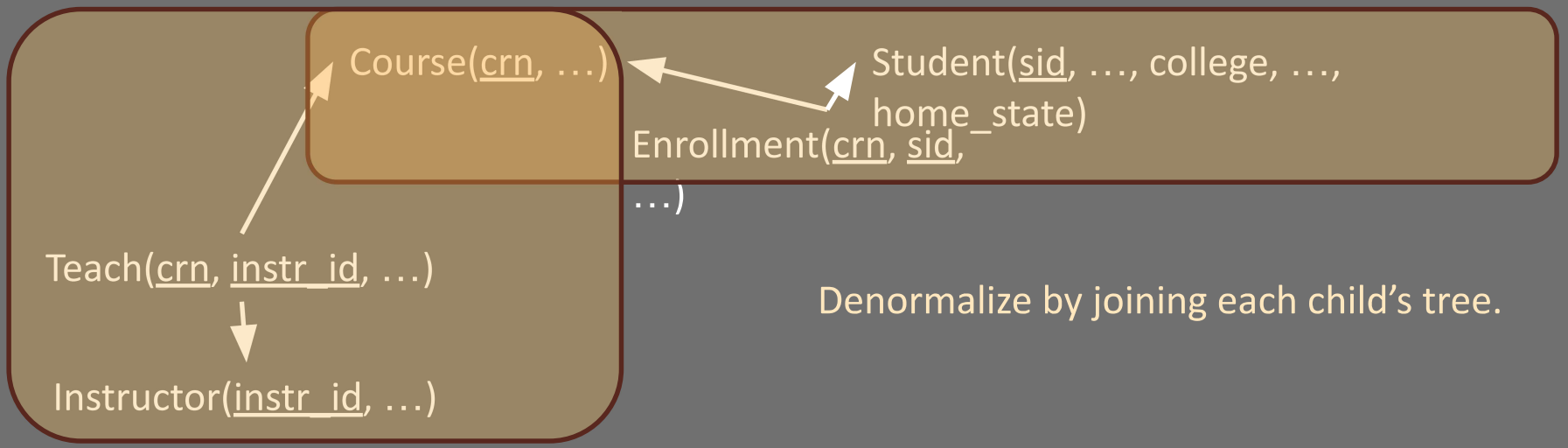


Typical:

- Few cycles.
- Super/sub-classes implemented only with superclass table.

Compressing starflakes to stars

Star = junction table & one level of child tables in 1-to-many relationships

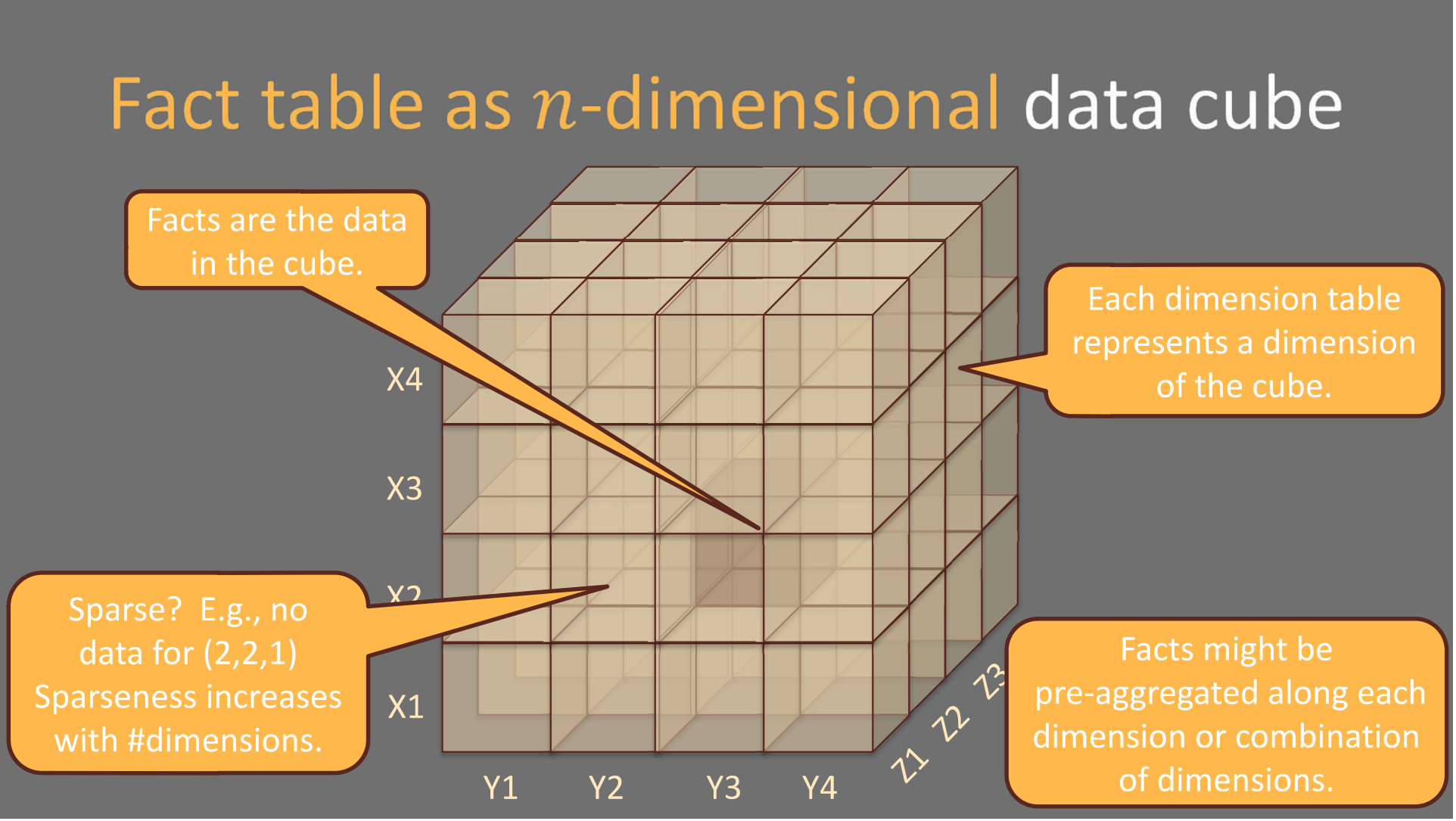


Denormalize by joining each child's tree.

ER-based dimensional modeling

- Collapse ER model into stars.
  - Stars preferred over starflakes unless #dimensions is very high.
  - Junction tables become fact tables.
  - Combine or break apart tables to group facts & dimensions.
- Make all keys synthetic.
- Add time dimension, if needed.
- Denormalize. Duplicate or pre-aggregate data to save query time.

Fact table as n-dimensional data cube



Sometimes things are still messy

Not all data fit nicely into facts + 1-to-many dimensions.

Leads to exceptions from this simple presentation.

Implementation strategies

Relational – Implemented as previously outlined in course.





Multidimensional – Implement as a n-dimensional cube.

- Optimizations for sparse regions of the cube.



Hybrid of the relational & multidimensional.